

Sampling Theory

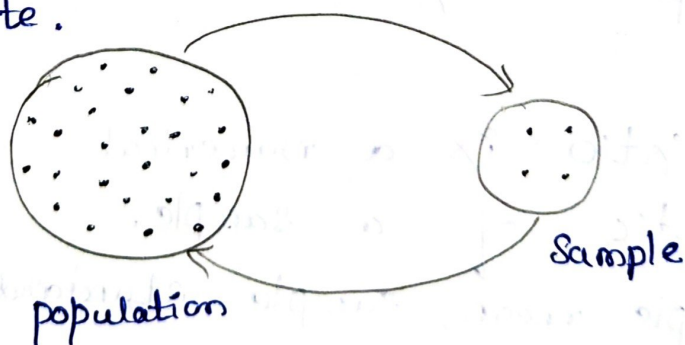
Sampling theory: population and sample, parameters and statistics, sampling distribution, standard error, testing of hypothesis, levels of significance, test of significance for large samples, comparison of large samples.

Introduction:-

Sampling is a statistical method of obtaining representative data (observations) from a group. we have been using sampling concepts in our day to day lives knowingly (or) unknowingly; for instance we take a handful rice to check the rice quality of the full lot. This is an example of random sampling from a large population.

Population (universe):-

The group of objects (individuals) under study is called population (or) universe. Universe may be finite (or) infinite.



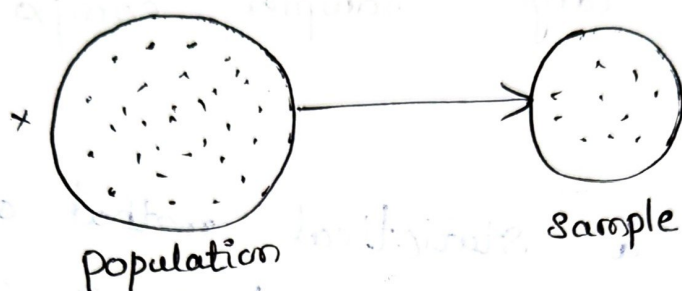
Sample:-

A part containing objects (individuals), selected from the population is called a sample.

Sample Size:-

(The number of individuals in a sample is called a Sample Size.)

Large and Small Sample
If the sample size 'n' is less than (or) equal to 30, then the sample is said to be small, otherwise it is called a large sample.



Parameters:-

✓ A parameter is a numerical value that describes a characteristic of a population.

Ex:- population mean, population standard deviation, population proportion.

In case if you wanted to know the average height of all adults in the world, that would be a population parameter.

Statistics:-

A statistic is a numerical value that describes a characteristic of a sample.

Ex:- Sample mean, sample standard deviation, sample proportion.

In case if you measured the height of a random group of 100 adults from different countries that would be a sample statistic.

Random Sampling

The selection of objects (individuals) from the universe in such a way that each object (individual) of the universe has the same chance of being selected is called Random sampling.

Lottery system is the most common example of random sampling.

Every random sampling need not be simple. For example, if balls are drawn without replacement from a bag of balls containing different balls, the probability of success changes in every trial. Thus, the sampling through random is not simple.

Simple Sampling :-

Simple sampling is a special case of random sampling in which each event has same probability of success (or) failure.

Sampling distribution :-

consider all possible samples of size n which can be drawn from a given population of random. For each sample, we can compute the mean. The mean of the samples will not be identical. If we group these different means according to their frequencies, the frequency distribution so formed is known as sampling distribution of the mean. Similarly we can have sampling distribution of the standard deviation etc.

Standard Error

The standard deviation of the sampling distribution is called the standard error (SE)

The reciprocal of the standard error is called precision.

Sampling Distribution of the means for ~~with~~ with / without replacement

1. Random Sampling with replacement

Here the items are drawn one by one and are put back to the population before the next draw.

If N is the size of the finite population and n is the size of the sample then we have " N^n " samples.

we have $\mu_{\bar{x}} = \mu$

$$\text{and } \sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}$$

where $\mu_{\bar{x}} \rightarrow$ sample mean

$\mu \rightarrow$ population mean

$\sigma_{\bar{x}}^2 \rightarrow$ variance of sample mean

$\sigma^2 \rightarrow$ variance of population

$\sigma_{\bar{x}}$ is also called the standard error.

2. Random Sampling without replacement

(3)

Here the items are drawn one by one and are not put back to the population before the next draw then there will be " $N C_n$ " samples.

we have $\mu_{\bar{x}} = \mu$

$$\text{and } \sigma_{\bar{x}}^2 = \left[\frac{N-n}{N-1} \right] \frac{\sigma^2}{n} = \frac{\sigma^2}{n}$$

where $c = \frac{N-n}{N-1}$ is called the finite population correction factor.

Note:-

● If N is very large, then c is closer to 1 and we have $\lim_{N \rightarrow \infty} \frac{N-n}{N-1} = 1$ and for N is large the correction factor can be omitted.

$$\text{Population Mean } (\mu) = \frac{\sum x}{n}$$

$$\text{Population Variance } (\sigma^2) = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

Example problems

1. If $\{1, 2, 3\}$ constitute a population. Find a mean and standard deviation for sampling distribution of the sample means in the case of
- (i) Random samples of size 2 with replacement.
 - (ii) Random samples of size 2 without replacement.

Soln:- Here N (size of the finite population) = 3

$$\text{population mean } (\mu) = \frac{1+2+3}{3} = 2$$

$$\text{population variance } (\sigma^2) = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

$$= \frac{1}{3} \{ (1-2)^2 + (2-2)^2 + (3-2)^2 \}$$

$$= \frac{1+0+1}{3} = \frac{2}{3}$$

case (i):- Random samples of size 2 ($n=2$) with replacement
The various possible samples are:

$(1, 1), (1, 2), (1, 3); (2, 1), (2, 2), (2, 3); (3, 1), (3, 2), (3, 3)$

There are $N^n = 3^2 = 9$ no. number.

The mean of these are 1, 1.5, 2, 1.5, 2, 2.5, 2, 2.5, 3

Now we prepare the frequency distribution of these means where x is the variate and f is the frequency.

x	1	1.5	2	2.5	3
f	1	2	3	2	1

we shall compute the mean and variance of this frequency distribution

$$\mu_{\bar{x}} = \frac{\sum fx}{\sum f} = \frac{1+3+6+5+3}{9} = 2$$

$$\sigma_{\bar{x}}^2 = \frac{\sum f(x - \mu_{\bar{x}})^2}{\sum f}$$

$$\sigma_{\bar{x}}^2 = \frac{1}{9} \{1(1-2)^2 + 2(1.5-2)^2 + 3(2-2)^2 + 2(2.5-2)^2 + 1(3-2)^2\}$$

$$\sigma_{\bar{x}}^2 = \frac{1}{9} (1+0.5+0+0.5+1) = \frac{1}{3}$$

Thus we have $\mu_{\bar{x}} = 2 = \mu$;

$$\sigma_{\bar{x}}^2 = \frac{1}{3} \text{ and}$$

$$\sigma_{\bar{x}}^2 = \frac{(2/3)}{2} = \frac{1}{3}$$

case(ii) :- Random sample of size 2 without replacement we have ${}^3C_2 = 3$ samples.

The three samples are (1,2), (2,3), (3,1)

The associated means are 1.5, 2.5, 2

Further we have,

$$\text{Mean} = \mu_{\bar{x}} = \frac{1.5+2.5+2}{3} = 2$$

$$\begin{aligned} \text{Variance} = \sigma_{\bar{x}}^2 &= \frac{1}{3} \{ (1.5-2)^2 + (2.5-2)^2 + (2-2)^2 \} \\ &= \frac{0.5}{3} = \frac{1}{6} \end{aligned}$$

Here we note that $\mu_x = \mu = 2$

$$\text{Also } \left[\frac{N-n}{N-1} \right] \frac{\sigma^2}{n} = \left[\frac{3-2}{3-1} \right] \frac{2/3}{2} = \frac{1}{6} = \sigma_x^2$$

$$\text{Thus } \sigma_x^2 = \left[\frac{N-n}{N-1} \right] \frac{\sigma^2}{n} = \frac{1}{6}$$

Hypothesis :-

A hypothesis is an assumption based on industrial evidences that leads itself to further testing and experimentation.

(OR)

In order to arrive at a decision regarding the population through a sample of the population we have to make certain assumption referred to as hypothesis which may (or) may not be true.

Null Hypothesis [H_0]:

The hypothesis formulated for the purpose of its rejection under the assumption that it is true is called "Null Hypothesis".

Alternative Hypothesis [H_1]:

Any hypothesis which is complimentary to the null hypothesis is called Alternative hypothesis.

Testing of Hypothesis :-

procedure which enable to decide whether to accept (or) reject a hypothesis are called testing of hypothesis.

Errors

In a test process there can be four possible situations of which two of the situations leads to the two types of errors and the same is presented as follows.

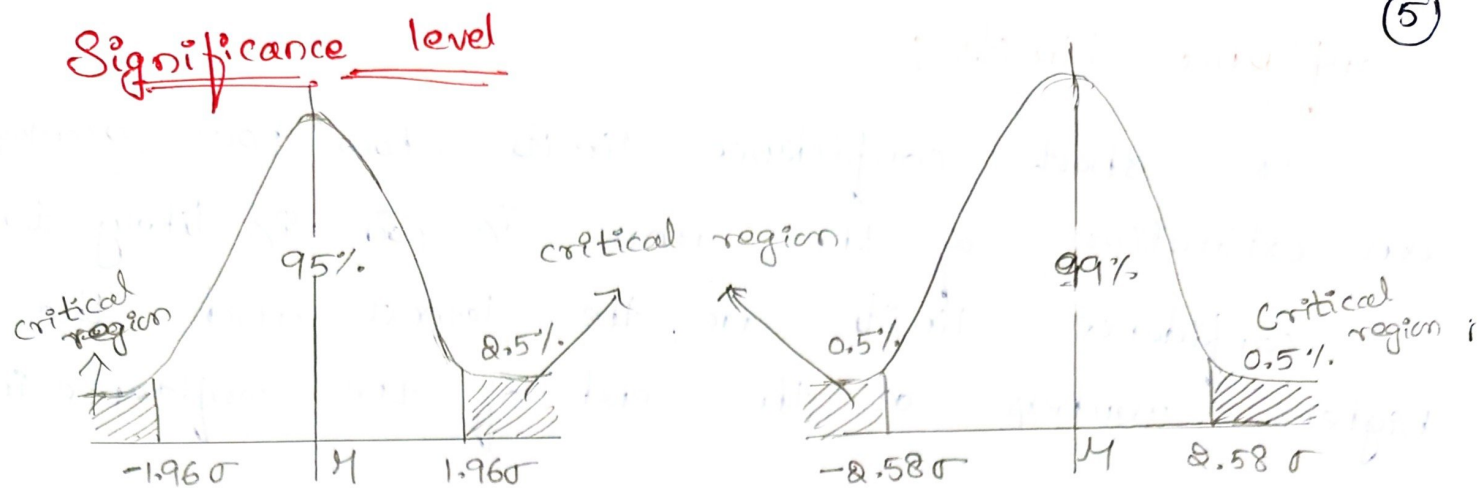
	Accepting the hypothesis	Rejecting the hypothesis
Hypothesis True	Correct decision	wrong decision (Type I error)
Hypothesis False	wrong decision (Type II error)	correct decision

Critical Region:-

A region which amounts to the rejection of null hypothesis is called critical region @ region of rejection. (OR)

A critical region, also known as the rejection region, is a set of values for the test statistic for which the null hypothesis is rejected.
ie if the observed test statistic is in the critical region that we reject the null hypothesis and accept the Alternative hypothesis

Significance level



The probability level, below which leads to the rejection of the hypothesis is known as the significance level @ level of significance.

This probability is conventionally fixed at 0.05 @ or 0.01 i.e. 5% @ or 1%.

The shaded portion in the figure corresponds to 5% and 1% level of significances.

Test of Significance

The process which helps us to decide about the acceptance @ rejection of the hypothesis is called the test of significance.

Confidence interval :-

A confidence interval is a range around a measurement that conveys how precise the measurement is. A confidence interval, in statistics, refers to the probability that a population parameter will fall between a set of values for a certain proportion of times. Analysis often use confidence intervals that contain either 95% @ or 99% of expected observations.

confidence Limits :-

In short, confidence limits show how accurate an estimation of the mean is. (a) is likely to be confidence limits are the lowest and the highest numbers at the end of the confidence interval.

Critical value

A critical value is the value of the test statistics which defines the upper and lower bounds of the confidence interval. (a) which defines the threshold level of statistical significance in a statistical test.

The region of significance will be of region to one side of the distribution. The area of such region will be equal to the level of significance itself. Such a test is called one tailed test.

The focus of attention lies on the value of Z on both sides of the mean i.e. lines in the two tails of the distribution then such a test is called two tailed test.

Test	critical values of Z	
	5% level	1% level
one-tailed test	-1.645 (or) 1.645	-2.23 (or) 2.33
Two-tailed test	-1.96 and 1.96	-2.58 and 2.58

① Test of significance of proportions

⑥

Standard normal variate $z = \frac{x - np}{\sqrt{npq}}$ (or)

$$Z = \frac{x - \mu}{\sigma}$$

Here p = probability of success

q = probability of failure

n = sample size

x = observed number of success

z = standard normal variate

μ = mean

σ = standard deviation.

1. A coin is tossed 1000 times and head turns up 540 times. Decide on the hypothesis that the coin is unbiased.

[unbiased = fair coin, Biased = ~~damaged~~ damaged coin]

$$\Rightarrow n = 1000$$

$$x = 540$$

p = probability of getting a head = $\frac{1}{2}$

$$p = \frac{1}{2}$$

$$p + q = 1 \Rightarrow q = 1 - p \Rightarrow \boxed{q = \frac{1}{2}}$$

$$Z = \frac{x - np}{\sqrt{npq}} = \frac{540 - 1000(0.5)}{\sqrt{1000 \times 0.5 \times 0.5}} = 2.53$$

$$Z = 2.53 < 2.58$$

Thus we can say that the coin is unbiased.

2. A 'die' is thrown 9000 times and a throw of 3 (or) 4 was observed 3240 times show that the die cannot be regarded as an unbiased one.

$$\rightarrow n = 9000$$

$$x = 3240$$

probability of getting a die of faces 3 (or) 4

$$p = \frac{2}{6} = \frac{1}{3}$$

$$p + q = 1$$

$$q = 1 - p = 1 - \frac{1}{3} = \frac{2}{3}$$

$$Z = \frac{x - np}{\sqrt{npq}} = \frac{3240 - 9000(\frac{1}{3})}{\sqrt{9000 \times \frac{1}{3} \times \frac{2}{3}}}$$

$$Z = 5.37$$

$$Z = 5.37 > 2.58$$

Thus we can say that the die is biased.

3. Result extract revealed that in a certain school over a period of five years 725 students had passed and 615 students had failed. Test the hypothesis that success and failure are in equal proportions.

$$\rightarrow n = 725 + 615 = 1340$$

$$x = 725$$

probability of success (passing) assuming equal proportions:

$$p = \frac{1}{2}$$

$$p + q = 1 \Rightarrow \boxed{q = \frac{1}{2}}$$

$$Z = \frac{x - np}{\sqrt{npq}} = \frac{725 - 1340(\frac{1}{2})}{\sqrt{1340 \times \frac{1}{2} \times \frac{1}{2}}} = 3.005$$

$$Z = 3.005 > 2.58$$

Thus, we can say that the success and failure are not in equal proportion. The hypothesis is rejected.

(7)

4. In 324 throws of a six face die and an odd no turned up 181 times, it is reasonable to think that the die is an unbiased one.

→ $n = 324$

$x = 181$

probability of getting odd number; $p = \frac{3}{6} = \frac{1}{2}$

$p + q = 1$

$q = 1 - \frac{1}{2} \Rightarrow \boxed{q = \frac{1}{2}}$

$$Z = \frac{x - np}{\sqrt{npq}} = \frac{181 - 324(\frac{1}{2})}{\sqrt{324 \times \frac{1}{2} \times \frac{1}{2}}} = 2.111$$

$Z = 2.111 < 2.58$

Thus we can say that die is unbiased.

Probable limits

1. $p \pm 2.58 \sqrt{pq/n}$ are the probable limits at 1% of significance level.

Standard Error = $\sqrt{\frac{pq}{n}}$

5. A sample of 900 days was taken in a coastal town and it was found that on 100 days. The weather was very hot. obtain the probable limits of the percentage of very hot weather.

→ probability of very hot weather = $p = \frac{100}{900} = \frac{1}{9}$

$n = 900$

$p + q = 1$

$q = 1 - \frac{1}{9} = \frac{8}{9}$

probable limits = $p \pm 2.58 \sqrt{\frac{pq}{n}}$

$= \frac{1}{9} \pm 2.58 \sqrt{\frac{1}{9} \times \frac{8}{9} \times \frac{1}{900}}$

$= 0.111 \pm 0.027$

$= 0.138 \text{ and } 0.084$

probable limits of very hot weather is 8.4% to 13.8%

6. In a sample of 500 men it was found that 60% of them had over weight. what can we infer about the proportion of people having over weight in the population.

→ probability of people having over weight

$$p = 60\% = \frac{60}{100} = 0.6$$

$$p + q = 1$$

$$q = 0.4$$

$$n = 500$$

$$\text{probable limits} = p \pm 2.58 \sqrt{\frac{pq}{n}}$$

$$= 0.6 \pm 2.58 \sqrt{\frac{0.6 \times 0.4}{500}}$$

$$= 0.6 \pm 0.0565$$

$$= 0.5435 \text{ and } 0.6565$$

probable limits of people having over weight is
54.35% and 65.65%.

7. A Survey was conducted in a slum locality of 2000 families by selecting a sample of size 800. It was revealed that 180 families were illiterate. Find the probable limits of the illiterate families in a population of 2000.

→ probability of illiterate families

$$p = \frac{180}{800} = 0.225$$

$$p + q = 1$$

$$q = 0.775$$

$$n = 800$$

$$\text{probable limits} = p \pm 2.58 \sqrt{\frac{pq}{n}}$$

$$= 0.225 \pm 2.58 \sqrt{\frac{0.225 \times 0.775}{800}}$$

$$= 0.225 \pm 0.038$$

$$= 0.187 \text{ and } 0.263$$

The probable limits of illiterate families in the population of 2000 is

$$\begin{array}{cc} 0.187 \times 2000 & 0.263 \times 2000 \\ 374 & 526 \end{array}$$

Thus 374 to 526 are probably illiterate families.

8. A random sample of 500 apples was taken from a large consignment and 65 were found to be bad. Estimate the proportion of bad apples in the consignment as well as the standard error of the estimate. Also find the percentage of bad apples in the consignment.

→ $n=500$
probability of bad $p = \frac{65}{500} = 0.13$

$$\begin{aligned} p+q &= 1 \\ q &= 1-0.13 \\ q &= 0.87 \end{aligned}$$

$$\begin{aligned} \text{probable limits} &= p \pm 2.58 \sqrt{\frac{pq}{n}} \\ &= 0.13 \pm 2.58 \sqrt{\frac{0.13 \times 0.87}{500}} \\ &= 0.13 \pm 0.038 \\ &= 0.092 \quad \text{and} \quad 0.168 \\ &= 9.2\% \quad \text{and} \quad 16.8\% \end{aligned}$$

$$\text{Standard Error} = \sqrt{\frac{pq}{n}} = \sqrt{\frac{0.13 \times 0.87}{500}} = 0.015$$

The required percentage of bad apples in the consignment lies between 9.20 and 16.87

2. Test of Significance of Difference between two Sample proportions

To test the significance of the difference between the samples, proportions, the test statistic under the null hypothesis H_0 that there is no significance difference between the two sample proportions we have

$$Z = \frac{P_1 - P_2}{\sqrt{PQ \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

* Sample proportions (P_1, P_2)

$$P_1 = \frac{x_1}{n_1}, \quad P_2 = \frac{x_2}{n_2}$$

x_1, x_2 = number of successes in each sample

n_1, n_2 = Sample Sizes.

* pooled proportion (P)

$$P = \frac{x_1 + x_2}{n_1 + n_2} \quad \text{or} \quad P = \frac{n_1 P_1 + n_2 P_2}{n_1 + n_2}$$

$$Q = 1 - P$$

* At 5% Significance : $Z_\alpha = 1.96$

* At 1% Significance : $Z_\alpha = 2.58$

Conclusion:-

* If $|Z| > Z_\alpha$: Reject the null hypothesis

* If $|Z| < Z_\alpha$: Accept the null hypothesis.

Problems

1. If a sample of 300 units of a manufactured product 65 units were found to be defective and in another sample of 200 units, there were 35 defectives. Is there significant difference in the proportion of defectives in the samples at 5% level of significance.

Given: $n_1 = 300$, $n_2 = 200$
 $x_1 = 65$, $x_2 = 35$

Null hypothesis: $H_0: P_1 = P_2$

Alternative hypothesis: $H_1: P_1 \neq P_2$

Level of significance $\alpha = 0.05$ (5%)

The sample of 300 units of a manufactured product 65 units were found to be defective

$$\therefore P_1 = \frac{65}{300} = 0.22$$

The sample of 200 units of a manufactured product 35 units were found to be defective

$$\therefore P_2 = \frac{35}{200} = 0.175$$

We know that
$$P = \frac{x_1 + x_2}{n_1 + n_2} = \frac{65 + 35}{300 + 200} = 0.2$$

$$P + Q = 1$$

$$Q = 1 - 0.2 = 0.8$$

$$Z = \frac{P_1 - P_2}{\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{0.22 - 0.175}{\sqrt{(0.2 \times 0.8)\left(\frac{1}{300} + \frac{1}{200}\right)}}$$

$$= \frac{0.045}{\sqrt{0.16 \times 0.0083}} = 1.235$$

$$Z = 1.235 < 1.96$$

There is no significant difference in the proportion of defectives in the two samples.

2. In a large city A, 20% of random sample of 900 school boys had a slight physical defect. In ~~900~~ other another large city B, 18.5% of a random sample, of 1600 school boys had the same defect. Is the difference between the proportions significant at 5% level of significance.

→ Given: $n_1 = 900$, $n_2 = 1600$

$x_1 = 20\%$ of random sample of 900 $= 0.2 \times 900 = 180$

$x_2 = 18.5\%$ of random sample of 1600 $= 0.185 \times 1600 = 296$

Null hypothesis $H_0: P_1 = P_2$

Alternative hypothesis $H_1: P_1 \neq P_2$

Level of significance $\alpha = 0.05$ (5%)

$$\therefore P_1 = \frac{180}{900} = 0.2$$

$$P_2 = \frac{296}{1600} = 0.185$$

we know that
$$p = \frac{x_1 + x_2}{n_1 + n_2} = \frac{180 + 296}{900 + 1600} = 0.1904$$

$$P + q = 1 \Rightarrow q = 0.8096$$

$$Z = \frac{P_1 - P_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{0.2 - 0.185}{\sqrt{(0.1904 \times 0.8096) \left(\frac{1}{900} + \frac{1}{1600} \right)}}$$

$$= \frac{0.0150}{\sqrt{0.1541 \times 0.00173}}$$

$$= \frac{0.0150}{\sqrt{0.000266}} = 0.9302$$

$$Z = 0.9302 < 1.96$$

$\therefore H_0$ is accepted

There is no significant difference in the proportion of boys with physical defects in the two cities.

3. In an exit poll enquiry it was revealed that 600 voters in one locality and 400 voters from another locality favoured 55% and 48% respectively a particular party to come to power. Test the hypothesis that there is a difference in the locality in respect of the opinion. Take (i) $\alpha = 0.05$
→ Given: $n_1 = 600$ $n_2 = 400$ (ii) $\alpha = 0.01$

$$x_1 = 55\% \text{ of } 600 \text{ voters} = 0.55 \times 600 = 330$$

$$x_2 = 48\% \text{ of } 400 \text{ voters} = 0.48 \times 400 = 192$$

Null hypothesis $H_0: P_1 = P_2$

Alternative hypothesis $H_1: P_1 \neq P_2$

Level of Significance $\alpha = 0.05$ and $\alpha = 0.01$

$$\therefore P_1 = \frac{330}{600} = 0.55$$

$$P_2 = \frac{192}{400} = 0.48$$

we know that

$$P = \frac{x_1 + x_2}{n_1 + n_2} = \frac{330 + 192}{600 + 400} = 0.522$$

$$P + q = 1$$

$$q = 0.478$$

$$Z = \frac{P_1 - P_2}{\sqrt{Pq\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{0.55 - 0.48}{\sqrt{(0.522)(0.478)\left(\frac{1}{600} + \frac{1}{400}\right)}}$$

$$= \frac{0.070}{0.03224} = 2.171$$

$$Z = 2.171 > 1.96, \quad Z = 2.171 < 2.58$$

Thus the null hypothesis that there is no difference between the localities is rejected at 5% level but not at 1% level of significance.

4. In a sample of 600 men from a certain city, 450 are found smokers. In other sample of 900 men from another city, 450 are smokers. Do the indicate that the cities are significantly different with respect to the habit of smoking among men. Test at 5% Significance level.

→ Given: $n_1 = 600$ $n_2 = 900$
 $x_1 = 450$ $x_2 = 450$

Null hypothesis $H_0: P_1 = P_2$

Alternative hypothesis $H_1: P_1 \neq P_2$

Level of Significance $\alpha = 0.05$

$\therefore P_1 = \frac{450}{600} = 0.75$ $P_2 = \frac{450}{900} = 0.5$

we know that $p = \frac{x_1 + x_2}{n_1 + n_2} = \frac{450 + 450}{600 + 900} = 0.6$

$$p + q = 1$$

$$q = 0.4$$

$$\begin{aligned} \therefore Z &= \frac{P_1 - P_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{0.75 - 0.5}{\sqrt{(0.6)(0.4) \left(\frac{1}{600} + \frac{1}{900} \right)}} \\ &= \frac{0.25}{0.02581} = 9.686 \end{aligned}$$

$$Z = 9.686 > 1.96$$

Thus the null hypothesis is reject.

Hence we conclude that there is a significant difference

5. One type of Aircraft is found to develop engine trouble in 5 flights out of a total of 100 and another type in 7 flights out of a total of 200 flights. Is there a significant difference in the two types of Aircrafts so far as engine defects are concerned. (5% level of significance)

→ Given: $n_1 = 100$ $n_2 = 200$
 $x_1 = 5$ $x_2 = 7$

Null hypothesis $H_0: P_1 = P_2$

Alternative hypothesis $H_1: P_1 \neq P_2$

Level of Significance: $\alpha = 0.05$

$\therefore P_1 = \frac{5}{100} = 0.05$ $P_2 = \frac{7}{200} = 0.035$

we know that

$$P = \frac{x_1 + x_2}{n_1 + n_2} = \frac{5 + 7}{100 + 200} = 0.04$$

$$P + Q = 1$$

$$Q = 0.96$$

$$Z = \frac{P_1 - P_2}{\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{0.05 - 0.035}{\sqrt{(0.04)(0.96)\left(\frac{1}{100} + \frac{1}{200}\right)}}$$

$$= \frac{0.015}{0.024} = 0.625$$

$$Z = 0.625 < 1.96$$

Thus the null hypothesis is accepted.

Hence we conclude that there is no significant difference.

6. Random sample of 1000 engineering students from a city A and 800 from city B were taken. It was found that 400 students in each of the sample were from payment quota. Does the data reveal a significant difference between the two cities in respect of payment quota students? Take (i) $\alpha = 0.05$ (ii) $\alpha = 0.01$

Given :- $n_1 = 1000$ $n_2 = 800$
 $x_1 = 400$ $x_2 = 400$

Null hypothesis $H_0: P_1 = P_2$

Alternative hypothesis $H_1: P_1 \neq P_2$

Level of significance $\alpha = 0.05$, $\alpha = 0.01$

$$\therefore P_1 = \frac{400}{1000} = 0.4 \quad P_2 = \frac{400}{800} = 0.5$$

we know that

$$p = \frac{x_1 + x_2}{n_1 + n_2} = \frac{400 + 400}{1000 + 800} = 0.444$$

$$p + q = 1$$

$$q = 0.555$$

$$z = \frac{P_1 - P_2}{\sqrt{pq\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{0.4 - 0.5}{\sqrt{(0.444)(0.555)\left(\frac{1}{1000} + \frac{1}{800}\right)}}$$

$$= \frac{-0.1}{0.0235} = -4.255$$

$$|z| = 4.255 > 1.96 \quad \& \quad |z| = 4.255 > 2.58$$

Thus the hypothesis H_0 is rejected both at 5% and 1% levels of significance.

③ Test of Significance for Single Mean:-

This test is used to find the significant difference between sample mean and population.

The test statistic is $|z| = \left(\frac{\bar{x} - \mu}{\sigma} \right) \sqrt{n}$

where $\bar{x}$ - sample mean

μ = population mean = μ

σ = standard deviation = $\sqrt{npq}$

1. A normal population has a mean of 6.8 and standard deviation of 1.5. A sample of 400 members gave a mean of 6.75. Is the difference significant. (5% level of significance)

Soln:- H_0 : There is no significant difference b/w $\bar{x}$ & μ

Given $\mu = 6.8$, $\sigma = 1.5$, $\bar{x} = 6.75$, $n = 400$

$$Z = \left(\frac{\bar{x} - \mu}{\sigma} \right) \sqrt{n}$$

$$Z = \left(\frac{6.75 - 6.8}{1.5} \right) \sqrt{400} = 0.67$$

$$|Z| = 0.67 < 1.96$$

$\therefore H_0$ is accepted.

$\therefore$ There is no significant difference b/w $\bar{x}$ & μ

2. A sample of 400 male students is found to have a mean height of 160 cms. Can it reasonably be regarded as a sample from a large population with mean height 162.5 cm and standard deviation 4.5 cms. (5% level of significance)

Soln:- H_0 : There is no significant difference $\bar{x}$ & μ

Given $\mu = 162.5$, $\sigma = 4.5$, $\bar{x} = 160$, $n = 400$

$$|Z| = \left(\frac{\bar{x} - \mu}{\sigma} \right) \sqrt{n} = \left(\frac{160 - 162.5}{4.5} \right) \sqrt{400}$$

$$|Z| = 11.1 > 1.96$$

$\therefore H_0$ is rejected

ie There is significant difference.

Problems

1. A manufacturer claimed that at least 95% of the equipment which he supplied to a factory conformed to specifications. An examination of a sample of 200 pieces of equipment revealed that 18 of them were faulty. Test his claim at a significance level of 1% & 5%.

→ Given $p=0.95$, $n=200$

hence $q=1-p=0.05$

$H_0: p=0.95$ & the claim is correct

$H_1: p \neq 0.95$ & the claim is false.

we choose the one tailed test

$$\mu = np = 200 \times 0.95 = 190$$

$$\sigma = \sqrt{npq} = \sqrt{200 \times 0.95 \times 0.05} = 3.082$$

Expected no of equipments according to the specification = 190
Actual no^r = 182 [$\because$ 18 out of 200 were faulty]

$$\therefore \text{difference} = 190 - 182 = 8$$

$$\text{Now } z = \frac{x - \mu}{\sqrt{npq}} = \frac{\text{difference}}{\sqrt{npq}} = \frac{8}{3.082} = 2.6$$

$$\therefore Z = 2.6 \begin{cases} > 1.645 = Z_{0.05} \\ > 2.33 = Z_{0.01} \end{cases}$$

Hence the given hypothesis is rejected at both 5% and 1% level of significance.

2. It has been found from experience that the mean breaking strength of a particular brand of thread is 275.6 gms with S.D of 39.7 gms. Recently a sample of 36 pieces of thread showed a mean breaking strength of 253.2 gms. can one conclude at a significance level of (a) 0.05 (b) 0.01 that the thread has become inferior?

→ $H_0: \mu = 275.6$ gms, mean breaking strength

$H_1: \mu \neq 275.6$ gms, inferior in breaking strength

we choose the one tailed test

Mean breaking strength of a sample of 36 piece = 253.2
difference = $275.6 - 253.2 = 22.4$

$$n = 36$$

$$Z = \frac{\text{difference}}{(\sigma/\sqrt{n})} = \frac{22.4}{(39.7/\sqrt{36})} = 3.38 \begin{cases} > Z_{0.05} = 1.645 \\ > Z_{0.01} = 2.33 \end{cases}$$

Thus the given hypothesis is rejected at 5% & 1% level of significance.

ie Under H_1 the thread has become inferior is accepted at both 5% & 1% level of significance.

3. In an examination given to students at a large no of different schools the mean grade was 74.5 and S.D grade was 8. At one particular school where 200 students took the examination the mean grade was 75.9. Discuss the significance of this result from the view point of (a) one tailed test (b) two tailed test at both 5% and 1% level of significance.

(15)

→ $H_0: \mu = 74.5$ and there is no change in the mean grade.
 $H_1: \mu \neq 74.5$ i.e. $\mu > 74.5$ and $\mu < 74.5$
 $\mu = 74.5$ and mean of a sample of size 200 (n)

$\therefore$ difference = $75.9 - 74.5 = 1.4$ is 75.9

$$z = \frac{\text{difference}}{(\sigma/\sqrt{n})} = \frac{1.4}{(8/\sqrt{200})} = 2.475$$

(a) one tailed test

$$\therefore z = 2.475 \begin{cases} > z_{0.05} = 1.645 \\ > z_{0.01} \end{cases}$$

at one tailed test the mean grade is significant at both 5% and 1% level.

(b) Two tailed test

$$z = 2.475 \begin{cases} > z_{0.05} = 1.96 \\ < z_{0.01} = 2.58 \end{cases}$$

at two tailed test the mean grade is significant at 5% level but not in 1% level.

Test of significance of difference b/w Mean

Let μ_1 & μ_2 be the mean of two population.

Let $(\bar{x}_1, \sigma_1)$; $(\bar{x}_2, \sigma_2)$ be the mean & S.D of two large samples of size n_1 & n_2 respectively. then the test statistic is given by

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$$

The confidence limits for the difference of means of the population are $(\bar{x}_1 - \bar{x}_2) \pm z_c \sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}$

→ $H_0: \mu = 74.5$ and there is no change in the mean grade.
 $H_1: \mu \neq 74.5$ i.e. $\mu > 74.5$ and $\mu < 74.5$
 $\mu = 74.5$ and mean of a sample of size 1200 (n)

$\therefore$ difference = $75.9 - 74.5 = 1.4$ i.e. 75.9

$$z = \frac{\text{difference}}{(s/\sqrt{n})} = \frac{1.4}{(8/\sqrt{1200})} = 2.475$$

(a) one tailed test

$$\therefore z = 2.475 \begin{cases} > z_{0.05} = 1.645 \\ > z_{0.01} \end{cases}$$

at one tailed test the mean grade is significant at both 5% and 1% level.

(b) Two tailed test

$$z = 2.475 \begin{cases} > z_{0.05} = 1.96 \\ < z_{0.01} = 2.58 \end{cases}$$

at two tailed test the mean grade is significant at 5% level but not in 1% level.

④ Test of Significance of difference b/w Mean

Let μ_1 & μ_2 be the mean of two population.

Let $(\bar{x}_1, \sigma_1)$; $(\bar{x}_2, \sigma_2)$ be the mean & S.D of two large samples of size n_1 & n_2 respectively. then the test statistic is given by

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$$

The confidence limits for the difference of means of the population are $(\bar{x}_1 - \bar{x}_2) \pm z_c \sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}$

Problems

1. In an elementary school examination the mean grade of 32 boys was 72 with a S.D of 8, while the mean grade of 36 girls was 75 with a S.D of 6. Test the hypothesis that the performance of girls are better than boys.

→ For boys $\bar{x}_B = 72$, $\sigma_B = 8$, $n_B = 32$

For girls $\bar{x}_G = 75$, $\sigma_G = 6$, $n_G = 36$

$$Z = \frac{\bar{x}_G - \bar{x}_B}{\sqrt{\frac{\sigma_G^2}{n_G} + \frac{\sigma_B^2}{n_B}}} = \frac{75 - 72}{\sqrt{\frac{36}{36} + \frac{64}{32}}} = 1.73 \begin{cases} > 1.645 = Z_{0.05} \\ < 2.33 = Z_{0.01} \end{cases}$$

∴ The difference in the performance of girls & boys in the examination is significant at 5% level but not in 1% level.

2. A sample of 100 bulbs produced by a company A showed a mean life of 1190 hours and a S.D of 90hrs. Also a sample of 75 bulbs produced by a company B showed a mean life of 1230 hours and a S.D of 120hrs. Is there a difference b/w the mean life time of the bulbs produced by the two companies at
(a) 5% level of significance
(b) 1% level of significance

→ Given, For company A: $\bar{x}_A = 1190$, $\sigma_A = 90$, $n_A = 100$

For company B: $\bar{x}_B = 1230$, $\sigma_B = 120$, $n_B = 75$

$$Z = \frac{\bar{x}_B - \bar{x}_A}{\sqrt{\frac{\sigma_B^2}{n_B} + \frac{\sigma_A^2}{n_A}}} = \frac{1230 - 1190}{\sqrt{\frac{(120)^2}{75} + \frac{(90)^2}{100}}} = 2.42 \begin{cases} > 1.645 = Z_{0.05} \\ > 2.33 = Z_{0.01} \end{cases}$$

The null hypothesis that there is no difference b/w the mean lifetime of bulbs is rejected at both 5% and 1% level of significance.

3. The mean of two large samples of 1000 & 2000 members are 168.75 cms and 170 cms respectively, can the samples be regarded as drawn from the same population of S.D 6.25 cms?

→ Given: $\bar{x}_1 = 168.75$, $\bar{x}_2 = 170$,
 $n_1 = 1000$, $n_2 = 2000$

$$Z = \frac{\bar{x}_2 - \bar{x}_1}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{170 - 168.75}{6.25 \sqrt{\frac{1}{1000} + \frac{1}{2000}}} = 5.16$$

$$\begin{cases} > 1.645 = Z_{0.05} \\ > 2.33 = Z_{0.01} \end{cases}$$

Thus the hypothesis is rejected at both 5% & 1% level of significance.

∴ The sample cannot be regarded as drawn from the same population.

4. A random sample for 1000 workers in a company has mean wage of Rs 50 per day and S.D of Rs 15. Another sample of 1500 workers from another company has mean wage of Rs 45 per day and S.D of Rs 20. Does the mean rate of wages varies b/w the two companies? Find the 95% confidence limits for the difference of the mean wages of the population of the two companies.

→ Given: Company-1: $\bar{x}_1 = 50$, $\sigma_1 = 15$, $n_1 = 1000$
 Company-2: $\bar{x}_2 = 45$, $\sigma_2 = 20$, $n_2 = 1500$

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{50 - 45}{\sqrt{\frac{(15)^2}{1000} + \frac{(20)^2}{1500}}} = 7.1306$$

$$\therefore Z = 7.1306 \begin{cases} > 1.645 = Z_{0.05} \\ > 2.33 = Z_{0.01} \end{cases}$$

Hence the given hypothesis is Rejected.

$\therefore$ The difference b/w the mean wages is not significant at both 5% & 1% level of significance.

$\therefore$ The 95% of confidence limits for the difference of mean wages is given by

$$(\bar{x}_1 - \bar{x}_2) \pm 1.96 \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

$$= 5 \pm 1.96 (0.7012)$$

$$= 5 \pm 1.374$$

$$= 3.63 \text{ and } 6.37$$

Thus we can say with 95% confidence that the difference of population mean of wages b/w the two companies lies b/w Rs. 3.63 and Rs. 6.37

Sampling Distribution of Mean

Standard normal variate $z = \frac{x - \mu}{\sigma}$

Problems

1. The weights of 1500 ball bearings are normally distributed with a mean of 635 gms. and S.D of 1.36 gms. If 300 random samples of size 36 are drawn from this population, determine the expected mean and S.D of the sampling distribution of means if sampling is done.

- (a) with replacement
(b) without replacement.

→ Given $N = 1500$ $\sigma = 1.36$
 $\mu = 635$ $n = 36$

(a) Expected mean $\mu_{\bar{x}} = \mu = 635$

$$\text{Expected S.D } \sigma_{\bar{x}} = \sqrt{\frac{\sigma^2}{n}} = \frac{\sigma}{\sqrt{n}} = \frac{1.36}{\sqrt{36}} = 0.227$$

(b) Expected mean $\mu_{\bar{x}} = \mu = 635$

$$\begin{aligned} \text{Expected variance } \sigma_{\bar{x}}^2 &= \frac{\sigma^2}{n} \left[\frac{N-n}{N-1} \right] \\ &= \frac{(1.36)^2}{36} \left[\frac{1500-36}{1500-1} \right] = 0.05 \end{aligned}$$

$$\text{Thus Expected S.D } \sigma_{\bar{x}} = \sqrt{0.05} = \underline{\underline{0.224}}$$

2. The weights of 1500 ball bearings are normally distributed with mean of 635 gms and S.D of 1.36 gms. If 300 random samples of size 36 are drawn from this population. In this case of random sampling with replacement, find how many random samples would have their mean

(a) between 634.76 gms and 635.24 gms

(b) ~~less~~ ^{greater} than 635.5 gms

(c) less than 634.2 gms

(d) less than 634.5 gms (or) more than 635.24 gms

Soln: Assume that the population is normal population and hence the normal distribution of means is also taken to be distributed normally.

The standard normal variate $z = \frac{x - \mu}{\sigma}$

Here $\mu_{\bar{x}} = 635$
 $\sigma_{\bar{x}} = 0.227$ [Refer last problem]

$$z = \frac{\bar{x} - \mu_{\bar{x}}}{\sigma_{\bar{x}}} = \frac{\bar{x} - 635}{0.227} \longrightarrow (1)$$

(a) probability of sample having mean b/w 634.76 and 635.24 is represented by

$$P(634.76 < \bar{x} < 635.24)$$

$$\text{from (1) if } \bar{x} = 634.76 \Rightarrow z = \frac{634.76 - 635}{0.227}$$

$$\text{if } \bar{x} = 635.24 \Rightarrow z = \frac{635.24 - 635}{0.227}$$

$$\begin{aligned} \therefore P(634.76 < \bar{x} < 635.24) &= P(-1.06 < z < 1.06) \\ &= 2P(0 < z < 1.06) \\ &= 2\phi(1.06) \\ &= 2(0.3554) \\ &= 0.7108 \end{aligned}$$

Thus for 300 samples the expected no of samples having their mean b/w 634.76 gms and 635.24 gms is given by $300 \times 0.7108 = 213.24 \approx 213$ samples.

(b) To find $P(\bar{x} > 635.5) \times 300$

by ① $\Rightarrow$ If $\bar{x} = 635.5$ then $z = \frac{635.5 - 635}{0.227} = 2.203$

$$\therefore P(\bar{x} > 635.5) = P(z > 2.203)$$

$$\therefore P(z > 2.203) = P(z > 0) - P(0 < z < 2.203)$$

$$= 0.5 - \phi(2.2)$$

$$= 0.5 - 0.4861$$

$$= 0.0139$$

Thus $P(\bar{x} > 635.5) \times 300 = 0.0139 \times 300 = 4.17 = 4$ samples.

(c) To find $P(\bar{x} < 634.2) \times 300$

by ① $\Rightarrow$ If $\bar{x} = 634.2$ then $z = \frac{634.2 - 635}{0.227} = -3.52$

$$\therefore P(\bar{x} < 634.2) = P(z < -3.52) = P(z > 3.52)$$

$$\therefore P(z > 3.52) = P(z > 0) - P(0 < z < 3.52)$$

$$= 0.5 - \phi(3.52)$$

$$= 0.5 - 0.4998 = 0.0002$$

Thus $P(\bar{x} < 634.2) \times 300 = 0.06 \approx 0$ samples.

(d) To find $[P(\bar{x} < 634.5) + P(\bar{x} > 635.24)] \times 300$

by ① $\Rightarrow$ If $\bar{x} = 634.5$ then $z = -2.2$

If $\bar{x} = 635.24$ then $z = 1.06$

$$\therefore P(\bar{x} < 634.5) + P(\bar{x} > 635.24) = P(z < -2.2) + P(z > 1.06)$$

$$\therefore P(z < -2.2) + P(z > 1.06) = P(z > 2.2) + P(z > 1.06)$$

$$= [P(z > 0) - P(0 < z < 2.2)] + [P(z > 0) - P(0 < z < 1.06)]$$

$$= [0.5 - \phi(2.2)] + [0.5 - \phi(1.06)]$$

$$= 1 - [\phi(2.2) + \phi(1.06)]$$

$$= 1 - [0.4861 + 0.3554]$$

$$= 0.1585$$

Thus $[P(\bar{x} < 634.5) + P(\bar{x} > 635.24)] \times 300 = 0.1585 \times 300$

$$= 47.55$$

$$\approx 48 \text{ samples}$$

3. Certain tubes manufactured by a company have mean life time of 800 hours and S.D of 60 hours. Find the probability that a random sample of 16 tubes taken from the group will have a mean life time.

(a) b/w 790 hrs and 810 hrs

(b) less than 785 hrs

(c) more than 820 hrs

(d) b/w 770 hrs and 830 hrs.

→ Given $\mu = 800$

$\sigma = 60$

$n = 16$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{60}{4} = 15$$

$$Z = \frac{\bar{x} - \mu}{\sigma_{\bar{x}}} = \frac{\bar{x} - 800}{15} \rightarrow \textcircled{1}$$

(a) To find $P(790 < \bar{x} < 810)$

by $\textcircled{1}$ If $\bar{x} = 790$ then $z = -0.67$

If $\bar{x} = 810$ then $z = 0.67$

$$\therefore P(-0.67 < z < 0.67) = 2P(0 < z < 0.67)$$

$$= 2\phi(0.67)$$

$$= 2(0.2486)$$

$$= 0.4972$$

$$\text{Thus } P(790 < \bar{x} < 810) = 0.4972$$

(b) To find $P(\bar{x} < 785)$

by $\textcircled{1}$ If $\bar{x} = 785$ then $z = -1$

$$\therefore P(z < -1) = P(z > 1) = P(z > 0) - P(0 < z < 1)$$

$$= 0.5 - \phi(1)$$

$$= 0.5 - 0.3413 = 0.1587$$

$$\text{Thus } P(\bar{x} < 785) = 0.1587$$

(c) To find $P(\bar{x} > 820)$

by $\textcircled{1}$ If $\bar{x} = 820$ then $z = 1.33$

$$P(z > 1.33) = P(z > 0) - P(0 < z < 1.33)$$

$$= 0.5 - \phi(1.33) = 0.5 - 0.4082 = 0.0918$$

$$\text{Thus } P(\bar{x} > 820) = 0.0918$$

(d) To find $P(770 < \bar{x} < 830)$

by ① if $\bar{x} = 770$ then $z = -2$

if $\bar{x} = 830$ then $z = 2$

$$\begin{aligned}\therefore P(-2 < z < 2) &= 2 P(0 < z < 2) = 2 \phi(2) \\ &= 2 (0.4772) = 0.9544\end{aligned}$$

Thus $P(770 < \bar{x} < 830) = 0.9544$